

CTD Post Processing Data Report (NABOS 2025)

Metadata

To check the metadata, we spot-checked individual CNVs to ensure the NMEA datetime were equivalent to the system time and that they matched the cast logs. Latitude and longitude stations were checked by plotting spatially.

Temperature

We encountered issues with secondary temperature on Station 41 – it appears that the pump was clogged (Figure 1). Secondary salinity values for Station 41 were converted to NAs during further data processing. No other issues were identified.

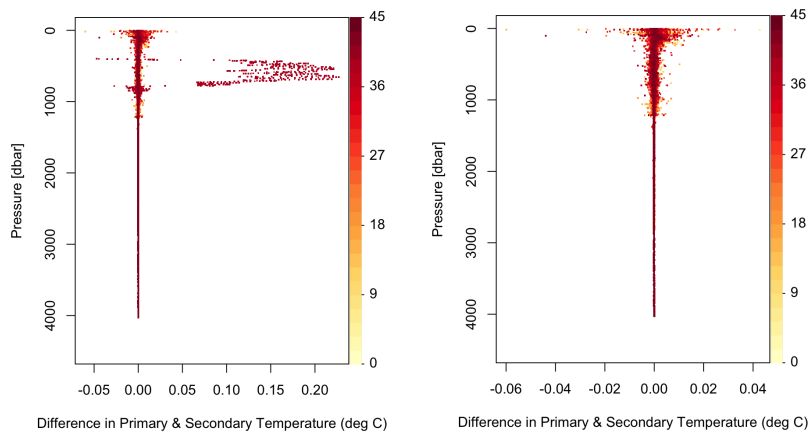


Figure 1. (Right) The difference in primary and secondary temperature prior to removal of station 41. (Left) The difference in primary and secondary temperature after removal of station 41. The color bar on both plots is station number.

Similar to the 2023 dataset: we note some large fluctuations in temperature (primary and secondary sensors) at the depth range of Barents Sea Branch Water (~600 – 1500 m) that may represent intrusions of water from the shelves or mixing between other water types. Throughout this sensor QC, we cannot find any indication that these features are due to sensor failure.

The primary temperature data is reported in the final QC'd downcast and bottle file.

Commented [LKI]: Is this accurate for 2025?

Conductivity

We encountered issues with secondary temperature on Station 41 – it appears that the pump was clogged. Secondary salinity values for Station 41 were converted to NAs during further data processing. No other issues were identified.

Similar to the 2023 dataset: we note some large fluctuations in temperature (primary and secondary sensors) at the depth range of Barents Sea Branch Water (~600 – 1500 m) that may represent intrusions of water from the shelves or mixing between other water types. Throughout this sensor QC, we cannot find any indication that these features are due to sensor failure.

Commented [LK2]: Is this accurate for 2025?

The primary conductivity sensor data is reported in the final QC'd downcast and bottle file.

Salinity

We encountered issues with secondary temperature on Station 41 – it appears that the pump was clogged. Secondary salinity values for Station 41 were converted to NAs during further data processing. No other issues were identified.

The primary salinity data (derived from secondary conductivity + secondary temperature sensors) is reported in the final QC'd downcast and bottle file.

Oxygen

The CTD Oxygen show large differences between primary and secondary sensors for a number of stations. Looking at cast-specific profile data, secondary sensors have consistent high variability (e.g., Fig X). No issues were observed with the primary oxygen sensor.

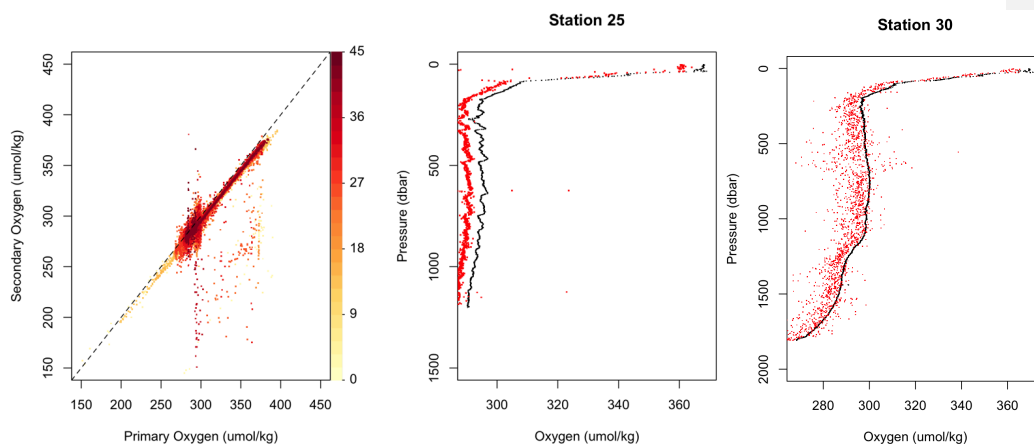


Figure 2. (Left) Primary vs Secondary Oxygen; the color bar is station number. (Center) Station 25 primary (black) and secondary (red) oxygen profiles. (Right) Station 30 primary (black) and secondary (red) profiles. Secondary center has more scatter throughout all stations.

A linear regression with the Winkler calibration samples was then conducted between the Winkler vs. Downcast Oxygen, for the primary sensor (Fig 3). There is a roughly 3% offset between the original downcast data and the corrected downcast.

The corrected CTD oxygen data from the primary oxygen sensor is reported in the final QC'd downcast and bottle file.

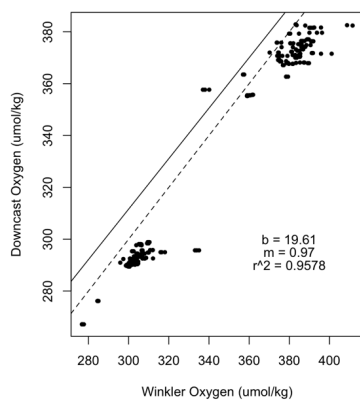


Figure 3. Winkler Titration oxygen concentration vs downcast, primary sensor oxygen. A correction to the downcast data is made using the linear regression of the data.

Chl Fluorometer

Chlorophyll Fluorescence is processed using SBE processing tools and the calibration provided by WetLabs. The data are not calibrated to discrete chlorophyll samples.

CDOM Fluorometer

CDOM Fluorescence is processed using SBE processing tools and the calibration provided by WetLabs. The data are not calibrated to discrete CDOM samples at this point. Contact Colin Stedmon for use (see data contacts section below).

Altimeter

No issues with the altimetry data is reported in this dataset. All values prior to activation of the sensor (100 m from the seafloor) are reported as NA.

Pressure

Pressure was checked by ensuring that casts have increasing pressure with time; there were no issues noticed in this dataset.

Other CTD Operational Notes

The CTD package stopped communicating on the downcast of station 14 (around 1000 m); it began communicating again at 1200 m. Currently, that gap has not been addressed, though future versions may pull in the upcast data to ensure a complete profile.

Other CTD Data Processing Notes

All changes made to the downcast data are applied to the bottle file as well. Downcast profiles for Temperature, Conductivity, Salinity, and Oxygen were investigated station-by station and any inconsistent data was flagged; only the oxygen data necessitated any flags. Flags used are 2 = good, 3 = questionable, and 4 = bad. Stations 43, 39, 35, 34, and 30 had some spikiness in the oxygen profiles that received 3 and 4 flags. All other data are flagged 2. Fluorescence and Transmissometry data are not flagged.

Data Contacts

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